

# A short-word obstruction for palindromic permutation blocks

Exact finite certificate; restricted construction family

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For a permutation  $\rho$  of five letters write  $P_\rho = \rho\rho^{\text{rev}}$ . We consider only words  $P_{\rho_0} \cdots P_{\rho_{q-1}}$ , having  $2q$  faces on each die. The following statement does not exclude arbitrary fair five-die words with the same face counts.

**Theorem 1.** *If a concatenation of five-letter palindromic permutation blocks is four-wise permutation-fair, then it contains at least twelve blocks. In particular this construction cannot give four-wise fairness with 18 faces on each die.*

*Proof.* Let  $D_3(\rho, p) = 3\#_p(P_\rho) - 4$  for ordered triples, and let  $D_4(\rho, p) = 3\#_p(P_\rho) - 2$  for ordered quadruples. Each palindrome block is pairwise fair. Products of two errors have degree at least six, so three times the degree-four error of the concatenation is the sum of vectors

$$C_{i,\rho}(p) = D_4(\rho, p) + 2iD_3(\rho, p_{2:4}) + 2(q-1-i)D_3(\rho, p_{1:3}).$$

Global relabeling lets us set  $\rho_0 = 01234$ .

The accompanying file `fourwise_palindrome_9_separator.json` gives a vector  $\lambda \in \{-1, 0, 1\}^{120}$  indexed by the lexicographically ordered quadruples. Direct integer enumeration of the 120 permutations verifies that

$$\langle \lambda, D_3(\rho, p_{2:4}) \rangle = 0, \quad \langle \lambda, D_4(\rho, p) \rangle = \langle \lambda, D_3(\rho, p_{1:3}) \rangle =: b_\rho,$$

where  $b_{01234} = 84$  and  $\min_\rho b_\rho = -18$ . Consequently, for any choices of the later blocks,

$$\left\langle \lambda, \sum_{i=0}^{q-1} C_{i,\rho_i} \right\rangle \geq 84(2q-1) - 18 \sum_{j=0}^{q-2} (2j+1) = 84(2q-1) - 18(q-1)^2.$$

This is strictly positive for  $1 \leq q \leq 10$ , contradicting four-wise fairness. Triple fairness also requires  $3 \mid q$ : the number of times each member of any fixed triple is first in the permutation blocks must be  $q/3$ . Thus  $q \geq 12$ .

The standalone verifier `verify_fourwise_palindrome_obstruction.py` uses only Python integers and no optimization package. The original search used an LP, but validity of the displayed certificate does not depend on numerical optimization.  $\square$

**Further finite searches.** The scripts `search_other_30_seeds.py` and `insert_fifth_into_30.py` test insertion of a fifth die in arbitrary gaps of already fair four-die, thirty-face words. Negative instances have explicit integral Farkas certificates. These are finite searches inside specified families, not a proof excluding general thirty-face five-die systems.